Gaurav C Kolte

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Solutions-driven Industrial Engineering graduate student with experience in data-driven process optimization, strong work ethic, rapid learning ability, and leadership in cross-functional teams. Seeking a **Spring 2026 internship/co-op** to contribute to a dynamic team for personal and organizational growth.

Education

Purdue University, West Lafayette, IN Jan 2025 - Dec 2026
Master of Science in Industrial Engineering
Coursework: Supply Chain Engineering, Statistical Quality Control, Linear programming, Product management

National Institute of Technology Tiruchirappalli, India July 2020 - May 2024
Bachelor of Technology in Production Engineering CGPA: 3.562/4

Experience

Graduate Research Assistant | Purdue University | West Lafayette May 2025 – Present
Next-generation Human-systems and Cognitive Engineering Lab, Purdue University

- Led the development of realistic simulation Scenarios in ISAT and MiniSim to evaluate automated flagger systems for the Indiana Department of Transportation (INDOT).
- Managed systematic data collection from human-subject testing for multiple flagger systems for performance comparison.
- Used JIRA to conduct and track 90+ simulation tests, making use of Kanban boards to schedule runs, monitoring progress to ensure effective data collection and timely reporting to cross-functional groups.

Industrial Engineering (Academic Project) | Aramark | West Lafayette Jan 2025 – May 2025

- Built a discrete-event simulation model in Arena to analyze customer flow and service efficiency for a Starbucks outlet serving 3,000+ customers per week.
- Collected and statistically fitted arrival and service time data (Exponential/Beta distributions) to model real-world variability across 7 resources and 2 customer order types.
- Performed bottleneck and capacity analysis using Arena's Process Analyzer, testing alternative staffing and equipment configurations over 10-day simulation runs
- Recommended adding an additional beverage machine, reducing peak wait times by 31% and improving overall service throughput by 10% during high-demand hours.
- Applied 5S Lean methodology to streamline workstation layout and implemented Poka – Yoke techniques such as colour coded ingredient stations and standard workflow to reduce preparation errors by 25%.

Business Analytics Intern | Sunrise Mentors Pvt Ltd (Coding Ninjas) | India Jan 2024 – Dec 2024

- Performed exploratory data analysis using MySQL and Excel for extraction, cleaning, and behaviour trend identification.
- Designed interactive Tableau dashboards to communicate complex technical data in easy-to-understand format, enabling stakeholders to quickly interpret retention KPIs .
- Mentored a team of 5 technical blog writers using JIRA to track progress and workflows, coordinating weekly output of 15 blogs with timely delivery and quality assurance.
- Collaborated cross-functionally with product, marketing, and operations teams to translate retention insights into process improvements and strategic initiatives.

Technical Skills

Technical: Python, SQL, Power BI, Tableau, Excel, ANSYS, Solidwork, SAP, AutoCAD, Catia, Minitab, Arena,
Industrial Methodologies\Tools: JIRA, JMP, Six Sigma, Lean Manufacturing, 7QC Tools, SPC
Supply Chain & Lean Tools: APQP, PPAP, 5S, Six Sigma, VSM

Technical Projects

Final Year Project – National Institute of Technology, Trichy | India Sept 2023 – May 2024

- Developed machine learning–based predictive maintenance (PdM) models using datasets from milling operations to forecast equipment failures and optimize maintenance scheduling.
- Evaluated 23 combinations of process parameters (air temperature, process temperature, torque, tool wear, rotational speed) using KNN, Random Forest, and SVM
- Achieved up to 96% prediction accuracy in estimating Remaining Useful Life (RUL) and identifying critical parameters affecting tool deterioration.
- Proposed an optimized PdM framework that reduced unplanned downtime by 30% and improved maintenance cost efficiency by 20%, contributing to sustainability and resource optimization.

Algorithm for solving linear programming problems using MATLAB. Aug 2024 - Dec 2024

- Developed a MATLAB code to solve LP problems which detects infeasibility, removes redundant constraints, prevents cycling.
- Formulated and solved programming models for 'XShipping Wood to Market' and 'Cereal Blending' problems
- Determined overall transport strategy to reduce total equivalent uniform annual cost while meeting investment budget and market sales demand for 'Shipping Wood to Market' problem.
- Identified the most profitable product mix that meets requirements of cereal blending company for 'Cereal Blending' problem.

Advanced Facility Planning project. Aug 2024 - Dec 2024

- Used Factory Cad for designing a facility showcasing the number of machines and workers required.
- Determined the Time standard and Takt time required for production of toolboxes annually.
- Projected the estimated budget required for the facility including office space and machine cost.
- Utilized ABC inventory system as cost effective method.

Leadership Experience

Head of Data Analytics, DATABYTE, The Data Science Club, NITT Sept 2023 – Jan 2024

- Delivered an Introduction to Data Science workshop for **150+** attendees.
- Led the development of a real-time Face Detection and Recognition Attendance System using python.
- Mentored 5 pre-final-year students, helping 100% secure internships/PPOs.

Placement Coordinator, Department of Training & Placement, NIT Trichy May 2023 – July 2024

- Facilitated campus placements and internships for over **1000** undergraduate students in the university by collaborating with the company's hiring representatives, managing student data, and attaining a stunning **95%** placements.

Manager, ProfNITT, Finance Club of NITT: Sept 2021 – Jan 2022

- Organized guest lectures on financial markets and career guidance, attracting 200+ attendees.
- Contacted more than 10 Industry veterans, converting 6 of them to share their expertise through guest lectures.

Head, Pragyan, ESI (Exhibition Sangam & Ingenium) Team Aug 2020-Mar 2024

- Led a 25+ member student team in organizing intra- and inter-college tech events and hackathons, overseeing event strategy, budgeting, logistics, and cross-team coordination.
- Mentored junior team members, reviewed event proposals and technical content, and facilitated collaboration with industry partners, building strengths in evaluation, stakeholder communication, and structured team guidance.

Research Publications

1.Food Supply Chain Resilience in the Digital Era: A Bibliometric Analysis and Development of Framework (**link**)
Journal of Business and Industrial Marketing, IF- 3.46

2.Food Supply Chain Disruptions Resilience: Framework and Review for Resilience Strategies in Digital Era. (**link**)
OPSEARCH Journal, Springer IF-1.6

3.A Case Study On Use Of Machine Learning Algorithms For Predictive Maintenance In Milling Operations. (**link**)
Next Research